

IBM Bob AI Hackathon 2026 · Track U1: Grid Reliability & Risk Intelligence

GridSentinel AI

Explainable composite risk intelligence for transformer fleets
– from SCADA telemetry to Granite-generated work orders.

Track U1 • Grid Reliability & Risk Intelligence

Presented by Team Mantra

THE PROBLEM

Hidden faults silently degrade transformers – until weather stress and blind spots turn a latent risk into a grid outage.

1

Hidden Transformer Faults

Dissolved gas anomalies and insulation degradation accumulate undetected inside transformers. SCADA telemetry alone cannot surface these latent faults – grid operators are flying blind until failure strikes.

2

Weather Compounding

Extreme heat, storms, and load surges accelerate fault progression. Without real-time weather-stress correlation, operators cannot anticipate when a marginal asset will cross the failure threshold.

3

Criticality Blind Spots

Not all transformers carry equal consequence. Utilities, transmission operators, and grid reliability engineers lack a composite risk score that weighs physical health, environmental stress, and grid criticality together.

OUR SOLUTION: END-TO-END PIPELINE

1



Ingest & Diagnose

SCADA/telemetry streams feed DGA physical health analytics – detecting dissolved gas anomalies and latent transformer faults in real time.

2



Stress & Criticality

Weather stress overlays and grid criticality mapping identify which assets face compounding environmental and topological risk.

3



Composite Risk Score

60/20/20 composite risk formula weights DGA health, weather stress, and grid criticality into a single explainable risk score per asset.

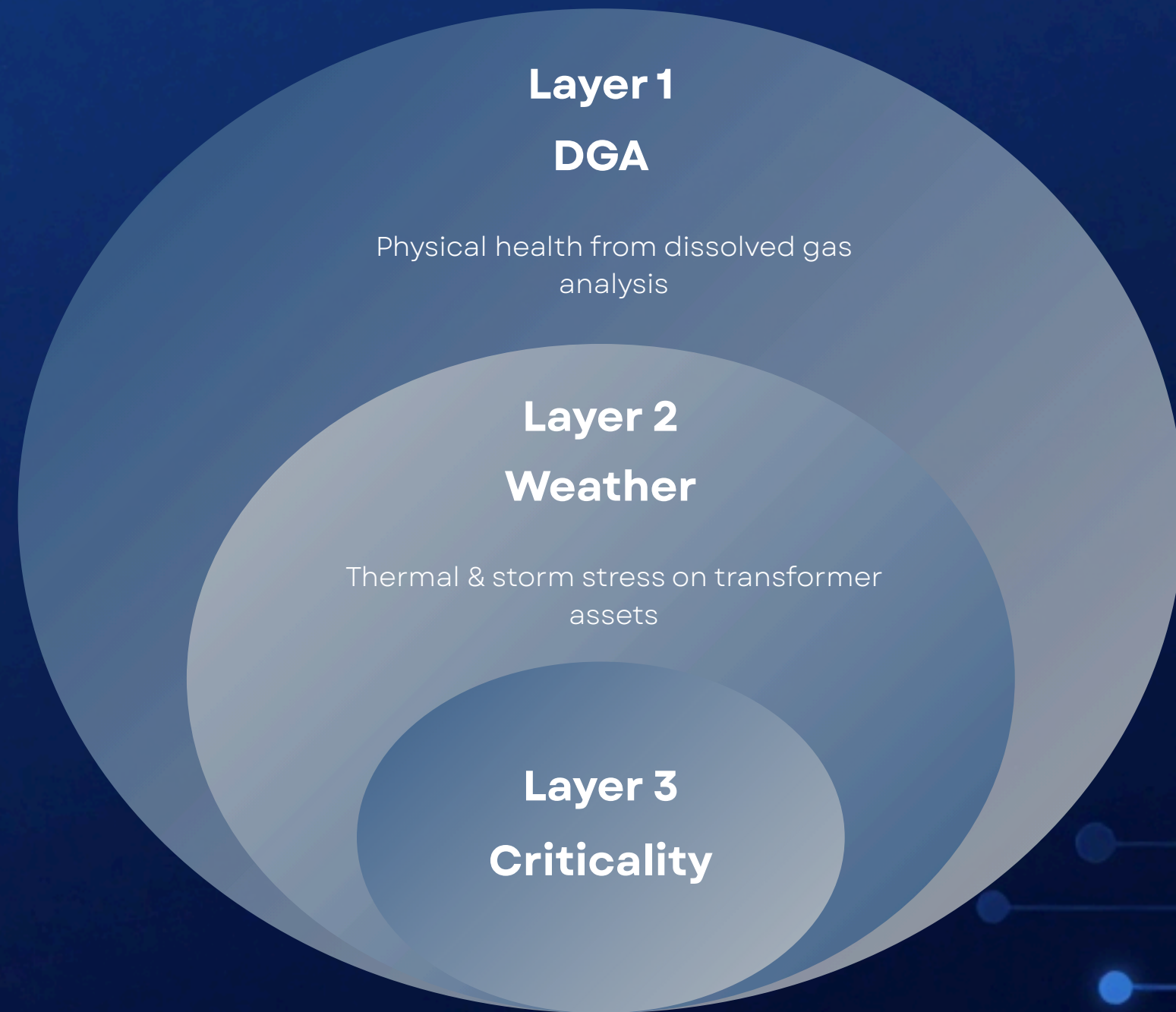
4



Granite Work Order

IBM Granite 3.0 synthesises a prioritised work order; a human operator countersigns before any field action – advisory only.

HOW IT WORKS: EXPLAINABLE RISK SCORING



60%

DGA Physical Health Score – dominant weight; dissolved gas ratios, fault gas trends, and anomaly severity from SCADA telemetry

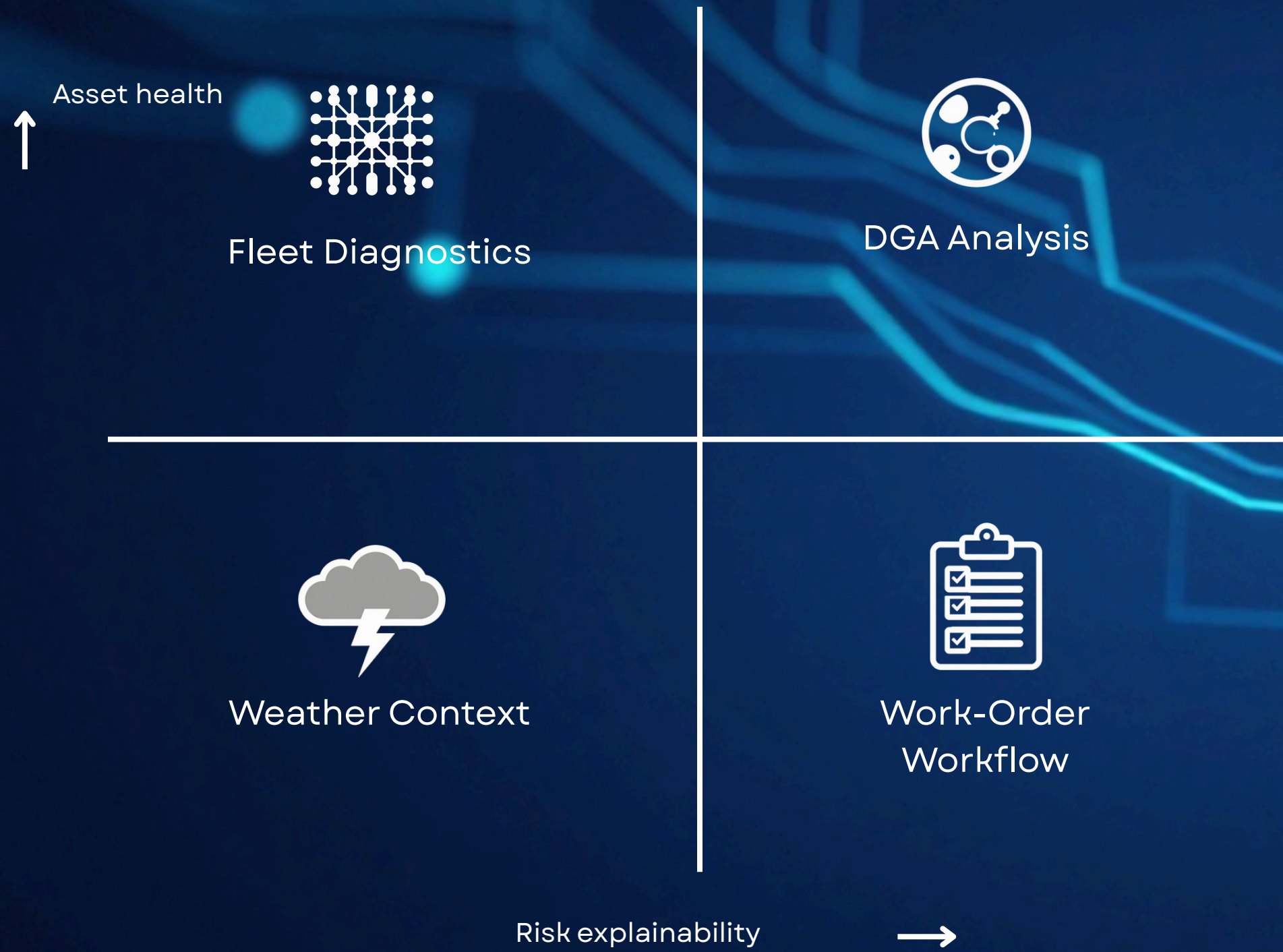
20%

Weather Stress Index – ambient temperature, humidity, and storm proximity amplifying thermal degradation risk

20%

Grid Criticality Score – load centrality, downstream dependency, and consequence severity if asset fails

PRODUCT EXPERIENCE



Command-Center UI

Fleet view flags high-risk transformers with composite score overlays

DGA diagnostics surface dissolved-gas trends and fault-type classification

Live weather stress layer shows heat, humidity and storm exposure per asset

Granite drafts prioritised work orders; engineer countersigns before dispatch

DECISION & SIMULATION: WHAT-IF PLANNING

1

Risk Explanation Engine

Breaks down every composite risk score into its 60/20/20 components – physical DGA health, weather stress, and grid criticality – with plain-language rationale for each flagged asset.

Fully explainable

2

Operational Contingency Simulator

Models cascading failure scenarios and rerouting options in a sandbox environment. Operators explore "what if this transformer fails?" without touching live SCADA data or control systems.

Read-only / no SCADA writes

3

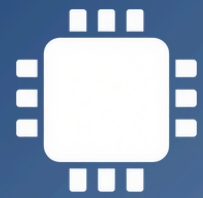
Operational Load Transfer Planner

Suggests safe load-transfer sequences to isolate high-risk assets during planned maintenance or emergency response, ranked by consequence severity and grid stability impact.

Advisory-only output

AI + RESPONSIBLE GOVERNANCE

GridSentinel AI is built on IBM Granite 3.0 with human-in-the-loop safeguards, offline resilience, and strict advisory-only boundaries.



Granite 3.0

IBM Granite 3.0 powers risk narrative generation and work-order synthesis. Developed entirely within IBM Bob IDE. Offline fallback mode ensures continuous scoring even when LLM connectivity is unavailable.



Human-in-Loop

Every AI-generated work order requires a human engineer countersign before dispatch. No automated action is taken on live SCADA. GridSentinel surfaces risk – operators make all final decisions.



Governance

GridSentinel operates in advisory-only mode with full audit trails. Risk scores are fully explainable via the 60/20/20 formula. No black-box outputs – every recommendation is traceable and contestable.

"Earlier attention to high-risk assets. Faster work-order synthesis. Consequence-aware prioritization – building a resilient, explainable grid."

GRIDSENTINEL AI – POWERING RESILIENT GRID OPERATIONS

Team Mantra – GridSentinel AI

Mahek Dhebariya

Team Lead & Frontend Architect

React 18 + Vite dashboard
animated gauges
FastAPI integration
UI/UX, presentation prep

Khushi Ka.Patel

Physics & Diagnostic Engine Lead

Authored 'src/dga_engine's py' per
IEEE C57.104
gas ratio and thermal degradation
algorithms

Palak Donga

Data, Criticality & IBM Granite Lead

Built 'criticality engine'
'weather engine'
engineered IBM Granite
prompt pipeline

Yashvi Antala

Quality Assurance & Testing Lead

Led system verification
test suite execution
documentation compliance